

Literature Benchmark Brief: Gonzalez_2019_RenewEnergy

Title: On the use of high-frequency SCADA data for improved wind turbine performance monitoring

Authors: E. Gonzalez, B. Stephen, D. Infield, J. J. Melero (2019)

Venue: Renewable Energy | **DOI:** 10.1016/j.renene.2018.10.019

Cluster: Cluster B: SCADA Degradation & Reliability

Core Theoretical Mechanism & Model Formulation

High-frequency vs 10-min SCADA comparison; sub-interval dynamics loss quantification

Empirical Scale & Experimental Dataset

Commercial wind turbines with synchronous high-speed recorders

Relevance to IEEE TSTE & Sustainable Energy Systems

Quantifies information loss when sub-10-min pitch transitions are collapsed into 10-min averages

Identified Gaps & Operational Vulnerabilities

Does not address multi-turbine spatial interactions or dispatch reserve impact

Role in Manuscript Defense & Reviewer #2 Proof

Supports Kelmarsh finding where 99.6% of pitch transitions occur between 10-min boundaries